close. This involved the aurora borealis – the Northern Lights – and their part in a sequence of events that could have led to thermonuclear war. For those wishing to anthropomorphise the natural world, it may have been the environment's attempt at revenge for global warming.

On Saturday 27 October, at 4.00am Eastern Daylight Time (EDT), a U-2 belonging to SAC's 4080th Strategic Wing took off from Eielson Air Force base in northern Alaska.[56] Major Charles ('Chuck') Maultsby's mission was to fly to the North Pole. His purpose was to collect radioactive samples from the atmosphere following several Soviet nuclear tests. Since 1958, the US, the USSR and the UK had been trying to negotiate a ban on nuclear testing and each observed a moratorium. This was abruptly ended by Moscow in September 1961 (despite Khrushchev's personal assurances to Kennedy that the USSR would not be the first to end the moratorium). Washington then resumed its own testing. The Soviets continued their programme during the missile crisis (as did the United States, which also kept test-firing ballistic missiles). SAC's High Altitude Sampling Program (HASP) missions also continued, even though as recently as 4 September 1962 a U-2 gathering electronic intelligence strayed off course into Soviet air space.

By measuring the HASP samples, it was possible to gauge the yield, and other characteristics, of the Soviet devices. In 1949 this was how it became known that Moscow had conducted an atomic test. Chuck Maultsby was one of the elite USAF pilots who flew the U-2 and was well-trained in operating an exceptionally challenging aircraft to fly. As he began to navigate his way across the Arctic Ocean, however, the aurora borealis was waiting for him.

Major Maultsby's mission was planned to last some eight hours. He had not previously flown to the North Pole, nor had he encountered the Northern Lights. To navigate his aircraft there and back, Maultsby had to fly beyond US radar coverage and could not rely on a magnetic compass. He was dependent on celestial navigation, using a sextant of a kind that mariners had depended upon for centuries, and which required him to know the position of various stars. The aurora borealis, however, prevented him from fixing his position. The U-2 flew a thousand miles off course and